

ABHIYAN SAINJU

Washington, DC

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Education

George Washington University

Master of Science - Computer Science — GPA: 4.0

January 2024 - December 2025

Washington, DC

St. Xavier's College

Bachelor of Science in Computer Science & Information Technology — GPA: 3.89

August 2018 - December 2022

Kathmandu, Nepal

Experience

George Washington University

August 2024 – Present

Graduate Teaching Assistant - Design and Analysis of Algorithms

Washington, DC

- Transformed algorithm comprehension for 60+ graduate students by designing interactive tutorials that translated complex concepts (dynamic programming, graph theory, NP-completeness) into industry case studies, resulting in 22% average improvement in midterm scores compared to previous cohorts.
- Managed comprehensive TA training program across 4 sections, personally onboarding 8 new TAs and standardized grading rubrics that reduced onboarding time from 2 weeks to 4 days while maintaining 95%+ grading consistency.
- Led curriculum redesign initiative with 3 faculty stakeholders, incorporating real-world optimization problems and managing \$2,400 educational resource budget that increased technical interview success rate by 18% among graduates.

ECS Tech

June 2023 – September 2023

Software Development Intern

Lalitpur, Nepal

- Developed and delivered 7 production-ready features (React.js, Node.js, PostgreSQL) for a fintech platform serving 1,200+ active users, including automated booking system that eliminated 89% of scheduling conflicts, saving customer support team 15 hours weekly.
- Owned end-to-end product launch of booking calendar feature, managing \$5,000 development budget and coordinating with 5 stakeholders (product, design, QA, customer success) to deliver 2 weeks ahead of schedule, directly contributing to 23% increase in customer retention.
- Led performance optimization initiative across React codebase, mentoring 2 junior interns on code-splitting techniques, achieving 43% load time improvement that boosted conversion rates from 12% to 16%.

Intel Security Service Pvt. Ltd.

August 2022 – March 2023

IT Solutions Lead - Contract

Kathmandu, Nepal

- Spearheaded complete digital transformation for security consultancy, rebuilding legacy website into lead-generation platform that generated 73% increase in qualified leads and \$35K in new contracts within first quarter.
- Managed \$12,000 annual IT budget as sole technical decision-maker, negotiating vendor contracts for hosting, security tools, and backup solutions while reducing operational costs by 40% and eliminating \$35K dependency on external contractors.
- Led client onboarding and technical training for 25-employee team, developing documentation and conducting 6 training sessions that reduced external support tickets by 80% and enabled internal team to handle 95% of routine maintenance independently.

Projects

InfraSight - Cloud Cost Intelligence Platform | Live Demo | GitHub

2025

Python, FastAPI, PostgreSQL, scikit-learn, Docker, React, Pandas

- Built a full-stack ML-powered cloud cost intelligence platform that cut AWS billing analysis time by 70%, enabling engineers and finance teams managing \$10K+ monthly budgets to detect inefficiencies faster than AWS Cost Explorer.
- Implemented forecasting, anomaly detection, and clustering pipelines (scikit-learn, Pandas), empowering DevOps and finance teams to catch cost spikes before billing cycles, preventing budget overruns.
- Implemented Random Forest + LSTM ensemble models achieving 92% accuracy in cost predictions, with containerized Docker deployment and JWT authentication demonstrating enterprise-ready architecture patterns.

MelodyHub - Real-Time Social Music Platform | GitHub

2025

Node.js, React, Socket.IO, Clerk Auth, Cloudinary, MongoDB

- Architected real-time music streaming platform supporting 100+ concurrent users with synchronized playback and live chat, achieving 99.2% uptime during 2-month beta testing with music communities totaling 50+ members.
- Led 3-person development team using Agile methodology, designing OOP-based backend architecture with inheritance patterns and comprehensive API documentation that reduced feature development time by 60%.
- Integrated Cloudinary CDN for audio streaming and Clerk authentication system, handling 500MB+ daily audio uploads while maintaining sub-3-second song loading times across different network conditions.

TalkifyDocs - AI SaaS Document Assistant | GitHub

2024

Next.js, OpenAI GPT-4, LangChain, Pinecone, Stripe, Prisma, TypeScript

- Built enterprise-ready AI document assistant that processes documents 47% faster than manual review, tested with 10 beta users from local research firms who reported significant time savings in contract analysis workflows.
- Managed \$150 monthly development budget for OpenAI API costs and hosting, while implementing Stripe integration and subscription billing system to demonstrate full-stack SaaS development capabilities.
- Engineered scalable RAG architecture using LangChain for contextual memory and Pinecone for semantic search, handling 10GB+ document repositories with sub-200ms query responses while maintaining 94% accuracy in document parsing.

Disease Prediction Using Machine Learning | GitHub
Python, scikit-learn, pandas, React, Tailwind CSS, Jupyter

2025

- Developed clinical decision support system using Random Forest on 4,920 medical records (131 symptoms, 41 disease classes), achieving 98.7% diagnostic accuracy that could assist healthcare providers in resource-limited settings.
- Built responsive diagnostic interface (React + Tailwind CSS) with symptom input workflows, designed for potential deployment in 3 rural clinics in Nepal as part of telemedicine initiative.
- Engineered robust preprocessing pipeline handling missing data and feature encoding, with planned SHAP integration for explainable AI to meet clinical transparency requirements for FDA approval pathway.

Multi-Source Information Retrieval Interface | GitHub
Python, GPT-3.5-turbo, Mistral 7B, FAISS, NLP

2024

- Built an NLP-based retrieval system integrating GPT-3.5 + FAISS that reduced research query times by 70%, enabling knowledge workers to quickly extract insights from PDFs, YouTube transcripts, and audio files.
- Expanded market reach by 40% through offline Mistral 7B integration, enabling privacy-sensitive organizations (law firms, hospitals) to process confidential documents without cloud dependencies.
- Authored comprehensive developer documentation and contribution guidelines, enabling 5 open-source contributors to add new AI models and features with average 2-day onboarding time.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frontend Frameworks: React.js, Next.js, Flutter, Tailwind CSS

Backend & APIs: Node.js, FastAPI, API Development, WordPress, Web Development Frameworks

Databases: PostgreSQL, MongoDB, MySQL

Cloud & DevOps: AWS (Cost Explorer, EC2, S3, IAM), Docker, Vercel

Machine Learning: scikit-learn, Pandas, Keras, TensorFlow, AI/ML Concepts, LangChain, FAISS, Pinecone

Tools & Methodologies: Git, Linux/Unix, Stripe, Agile, UML, Unit Testing, Data Structures & Algorithms, OOP, System Design